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Device with dielectric metal oxide layers and semiconductor apparatus including the same

Abstract

Provided are a semiconductor device and a semiconductor apparatus including the same, the semiconductor device including: a first electrode; a second electrode apart from the first electrode; a dielectric structure provided between the first electrode and the second electrode and including a dielectric layer including a metal oxide represented by $M_{\text{sub}}xO_{\text{sub}}y$; and a leakage current reducing layer including a metal oxide represented by $La_{\text{sub}}y'M'_{\text{sub}}y'O_{\text{sub}}z'$.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0111205, filed on Aug. 23, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) The disclosure relates to a semiconductor device and a semiconductor apparatus including the same.

2. Description of the Related Art

(3) As electronic devices are down-scaled, the available space to be occupied by electronic components, such as semiconductor devices, in electronic devices has also been reduced. Accordingly, along with the reduction in size of an electronic component and/or semiconductor device, such as a capacitor, a reduction in thickness of a dielectric layer of the capacitor is also required. However, in this case, a large leakage current may occur through the dielectric layer of the capacitor, thereby making it difficult to drive the electronic component and/or semiconductor device.

SUMMARY

(4) Provided are electronic components having a low leakage current and high capacitance, and semiconductor apparatuses including the same.

(5) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented example embodiments of the disclosure.

(6) According to an aspect of an embodiment, an electronic component includes: a first electrode; a second electrode apart from the first electrode; and a dielectric structure between the first electrode and the second electrode, the dielectric structure including a dielectric layer including a metal oxide represented by $MxOy$, where M is at least one of Ca, Sr, Ba, Sc, Y, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, and Lu, and a leakage current reducing layer including a metal oxide represented by $La.sub.xM'.sub.yO.sub.z'$, where M' is at least one of Ca, Sr, Ba, Sc, Y, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, and Lu).

(7) The dielectric layer may include first and second dielectric layers, and the leakage current reducing layer may be between the first dielectric layer and the second dielectric layer.

(8) The leakage current reducing layer may be between the first electrode and the dielectric layer.

(9) The leakage current reducing layer may be between the second electrode and the dielectric layer.

(10) A thickness of the leakage current reducing layer may be 0.1 Å or more and 4.5 Å or less.

(11) A total thickness of the dielectric structure may be 50 Å or less.

(12) The dielectric layer may have a single-layer structure or a multi-layer structure in which different materials are stacked.

(13) The first and second electrodes may each independently include a metal, a metal nitride, a metal oxide, or a combination thereof.

(14) One of the first and second electrodes may include a semiconductor material.

(15) According to an aspect of another embodiment, a semiconductor apparatus includes: a field effect transistor; and a capacitor electrically connected to the field effect transistor. The capacitor includes a first electrode; a second electrode apart from the first electrode; a dielectric structure between the first electrode and the second electrode, the dielectric structure including a dielectric layer including a metal oxide represented by $MxOy$, where M is at least one of Ca, Sr, Ba, Sc, Y, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, and Lu, and a leakage current reducing layer including a metal oxide represented by $La.sub.xM'.sub.yO.sub.z$, where M' is at least one of Ca, Sr, Ba, Sc, Y, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, and Lu.

- (16) The field effect transistor may include a channel layer between a source and a drain; a dielectric layer on the channel layer; and a gate electrode on the dielectric layer.
 - (17) The dielectric layer may include first and second dielectric layers, and the leakage current reducing layer may be between the first dielectric layer and the second dielectric layer.
 - (18) The leakage current reducing layer may be between the dielectric layer and at least one of the first electrode or the second electrode.
 - (19) A thickness of the leakage current reducing layer may be 0.1 Å or more and 4.5 Å or less.
 - (20) A total thickness of the dielectric structure may be 50 Å or less.
 - (21) The first and second electrodes may each independently include a metal, a metal nitride, a metal oxide, or a combination thereof.
 - (22) According to an aspect of another embodiment, an electronic device includes the above-described semiconductor apparatus.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features, and advantages of certain example embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a cross-sectional view of an electronic component according to some example embodiments;
- (3) FIG. 2 is a view illustrating comparison results of capacitance measurements of an electronic component including an Al.sub.xZr.sub.yO.sub.z leakage current reducing layer and an electronic component including a La.sub.xZr.sub.yO.sub.z leakage current reducing layer;
- (4) FIG. 3 is a view illustrating comparison results of measurement of leakage current values of an electronic component including an Al.sub.xZr.sub.yO.sub.z leakage current reducing layer and an electronic component including a La.sub.xZr.sub.yO.sub.z leakage current reducing layer;
- (5) FIG. 4 is a cross-sectional view of an electronic component according to some example embodiments;
- (6) FIG. 5 is a cross-sectional view of an electronic component according to some example embodiments;
- (7) FIG. 6 is a view of a semiconductor apparatus according to some example embodiments;
- (8) FIG. 7 is a view of a semiconductor apparatus according to some example embodiments;
- (9) FIG. 8 is a cross-sectional view of the semiconductor apparatus of FIG. 7 taken along line A-A';
- (10) FIG. 9 is a view of a semiconductor apparatus according to some example embodiments;
- (11) FIG. 10 is a conceptual diagram of a device architecture applicable to an electronic device according to some example embodiments; and
- (12) FIG. 11 is a conceptual diagram of a device architecture applicable to an electronic device according to some example embodiments.

DETAILED DESCRIPTION

(13) Reference will now be made in detail to some embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In the drawings, the sizes of constituent elements may be exaggerated for clarity. The embodiments described below are only examples, and thus, it should be understood that the embodiments may be modified in various forms. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the example embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements,

modify the entire list of elements and do not modify the individual elements of the list.

(14) Hereinafter, an expression such as “above” or “on” may include not only the meaning of “immediately on/under/to the left/to the right in a contact manner”, but also the meaning of “on/under/to the left/to the right in a non-contact manner”. Additionally, device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. In addition, it will be understood that when a unit is referred to as “comprising” another element, it does not preclude the possibility that one or more other elements may exist or may be added.

(15) When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing tolerance (e.g., $\pm 10\%$) around the stated numerical value. Further, regardless of whether numerical values or shapes are modified as “about” or “substantially,” it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical values or shapes.

(16) It will be understood that, although the terms “first,” “second,” “third”, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are used only to differentiate an element from another element, and the order and type of the elements are not limited thereto. Unless explicitly stated or contradicted to the order of operations constituting the method, these operations may be performed in an appropriate order and are not necessarily limited to the order described. In addition, the terms “-er”, “-or”, and “module” described in the specification mean units for processing at least one function and/or operation and can be implemented by hardware components, software components, and/or combinations thereof. For example, the processing units may include and/or be included in, but are not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), and programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc.

(17) The connecting lines, or connectors shown in the various figures presented are intended to represent example functional relationships and/or physical or logical couplings between the various elements. It should be noted that many alternative or additional functional relationships, physical connections or logical connections may be present in a practical device.

(18) The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the inventive concepts, and does not pose a limitation on the scope of the inventive concepts unless otherwise claimed.

(19) According to an aspect, an electronic component having a small leakage current and high capacitance may be provided.

(20) FIG. 1 is a cross-sectional view illustrating an electronic **100** according to some example embodiments. The electronic component **100** shown in FIG. 1 may be semiconductor device and/or a capacitor.

(21) Referring to FIG. 1, first and second electrodes **111** and **112** are provided to be apart from each other, and first and second dielectric layers **121** and **122** are provided on the first and second electrodes **111** and **112**, respectively. For example, the first electrode **111**, the first dielectric **121**, the second dielectric **122**, and the second electrode may be sequentially stacked. In addition, a leakage current reducing layer **130** is provided between the first and second dielectric layers **121** and **122**.

(22) The first electrode **111**, which may be referred to as a lower electrode, may be arranged on a substrate (not shown). The substrate may be a portion of a structure supporting the electronic component **100** (e.g., a capacitor) and/or a portion of a device connected to the electronic component **100**. The substrate may include a pattern of semiconductor material, a pattern of insulating material, and/or a pattern of conductive material. The substrate may include, for

example, a substrate **11**', a gate stack **12**, an interlayer insulating layer **15**, a contact structure **20**', and/or a bit line structure **13** in FIGS. **8** and **9** to be described later below.

(23) The substrate may include, for example, a semiconductor material such as at least one of silicon (Si), germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), indium phosphide (InP), and/or the like, and/or the substrate may include, for example, an insulating material such as at least one of silicon oxide, silicon nitride, silicon oxynitride, and/or the like.

(24) The second electrode **112**, which may be referred to as an upper electrode, may be arranged to face the first electrode **111** while being apart from each other. The first and second electrodes **111** and **112** may each independently include a conductive (and/or semiconductive) material, such as at least one of a metal, a metal nitride, a metal oxide, or a combination thereof. For example, the first and second electrodes **111** and **112** may each independently include a metal (such as ruthenium (Ru), titanium (Ti), tantalum (Ta), niobium (Nb), iridium (Ir), molybdenum (Mo), tungsten (W), platinum (Pt), and/or the like), a conductive metal nitride (such as titanium nitride (TiN), tantalum nitride (TaN), niobium nitride (NbN), molybdenum nitride (MoN), cobalt nitride (CoN), tungsten nitride (WN), and/or the like), and/or a conductive metal oxide (such as platinum oxide (PtO), iridium oxide (IrO.sub.2), ruthenium oxide (RuO.sub.2), strontium ruthenium oxide (SrRuO.sub.3), barium strontium ruthenium oxide ((Ba,Sr)RuO.sub.3), calcium ruthenium oxide (CaRuO.sub.3), lanthanum strontium cobalt oxide ((La,Sr)CoO.sub.3), and/or the like).

(25) For example, the first and second electrodes **111** and **112** may each independently include a metal nitride represented by MeMe'N, where Me is a metal element, Me' is an element different from Me, and N is nitrogen. This metal nitride may include MeN metal nitride doped with element Me'.

(26) Me may be, for example, one, two, or more elements selected from Be, B, Na, Mg, Al, Si, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, Cs, Ba, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, Fr, Ra, Ac, Th, Pa, and/or U.

(27) Me' may be, for example, one, two, or more elements selected from H, Li, Be, B, N, O, Na, Mg, Al, Si, P, S, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, As, Se, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, Cs, Ba, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, Fr, Ra, Ac, Th, Pa, and/or U.

(28) The first and second electrodes **111** and **112** may each independently include a single material layer and/or have a structure in which a plurality of material layers are stacked. For example, the first and/or second electrodes **111** and **112** may be a single layer of titanium nitride (TiN) and/or of niobium nitride (NbN). Alternatively, the first and/or second electrodes **111** and **112** may have a structure in which at least a first electrode layer including titanium nitride (TiN) and a second electrode layer including niobium nitride (NbN) are stacked.

(29) A first dielectric layer **121** is provided on a surface (e.g., an upper surface) of the first electrode **111**, and a second dielectric layer **122** is provided on a surface (e.g., a lower surface) of the second electrode **112**. The first and second dielectric layers **121** and **122** may include a dielectric material having paraelectric properties. For example, the first and second dielectric layers **121** and **122** may include a dielectric material having a dielectric constant of about 20 or more and/or 70 or less.

(30) In some example embodiments, each of the first and second dielectric layers **121** and **122** may include a metal oxide represented by M.sub.xO.sub.y (x and y are natural numbers), wherein O is oxygen and M may be one metal element selected from Ca, Sr, Ba, Sc, Y, La, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb and/or Lu. The first and/or the second dielectric layers **121** and **122** may include dopants. The first and second dielectric layers **121** and **122** may include, but are not limited to, the same metal oxide. For example, the first and second dielectric layers **121** and **122** may include the same and/or different metal oxides. The first and second dielectric layers **121** and **122**

may each independently have a single-layer structure including a single material layer or a multi-layer structure in which a plurality of material layers are stacked. The first and second dielectric layers may have the same and/or different thicknesses.

(31) The leakage current reducing layer **130** may be provided between the first and second dielectric layers **121** and **122**. The leakage current reducing layer **130** may reduce a leakage current flowing inside the electronic component **100** (e.g., a capacitor). In some example embodiments, the leakage current reducing layer **130** may include a metal oxide represented by

$\text{La.sub.xM'.sub.yO.sub.z'}$ (x' , y' , z' are natural numbers). M' may be one metal element selected from Ca, Sr, Ba, Sc, Y, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb and/or Lu. Such metal oxides may have paraelectric properties. The M in first and/or second dielectric **121** and **122** and the M' in the leakage current reducing layer **130** may be the same and/or different metal elements.

(32) The leakage current reducing layer **130** may have a thickness of about 0.1 Å or more and/or 4.5 Å or less. A total thickness of the first and second dielectric layers **121** and **122** and the leakage current reducing layer **130** may be about 50 Å or less. In some example embodiments, the total thickness of the first and second dielectric layers **121** and **122** and the leakage current reducing layer **130** may be about 40 Å or more and/or 50 Å or less. However, the disclosure is not limited thereto.

(33) In the electronic component **100** (e.g., a capacitor) according to some example embodiments, by providing the leakage current reducing layer **130** including a metal oxide represented by $\text{La.sub.xM.sub.yO.sub.z}$ between the first and second dielectric layers **121** and **122**, the capacitance may be increased, and a leakage current value may be decreased.

(34) In the above description, a case in which the electronic component **100** is a capacitor having a metal-insulator-metal (MIM) structure in which both the first and second electrodes **121** and **122** include a conductive material has been described. However, the present embodiment is not limited thereto. For example, the electronic component **100** may be/or include a semiconductor device such as a capacitor having a metal-insulator-semiconductor (MIS) structure in which one of the first and second electrodes includes a conductive material and the other includes a semiconductor material.

(35) FIGS. 2 and 3 show characteristics of a conventional electronic component including an $\text{Al.sub.xZr.sub.yO.sub.z}$ leakage current reducing layer and an electronic component according to some example embodiments including a $\text{La.sub.xZr.sub.yO.sub.z}$ leakage current reducing layer. In FIGS. 2 and 3, a capacitor having a MIM structure is used as the electronic component.

(36) FIG. 2 is a view illustrating comparison results of capacitance measurements of the conventional electronic component including an $\text{Al.sub.xZr.sub.yO.sub.z}$ leakage current reducing layer and the electronic component, according to some example embodiments, including a $\text{La.sub.xZr.sub.yO.sub.z}$ leakage current reducing layer. Referring to FIG. 2, it can be seen that the capacitance of the electronic component according to the example embodiments is improved by about 10% compared to the conventional electronic component.

(37) FIG. 3 is a view illustrating comparison results of measurement of leakage current values of a conventional electronic component including an $\text{Al.sub.xZr.sub.yO.sub.z}$ leakage current reducing layer and an electronic component, according to some example embodiments, including a $\text{La.sub.xZr.sub.yO.sub.z}$ leakage current reducing layer. Referring to FIG. 3, it can be seen that in the electronic component according to the example embodiments, a leakage current is reduced by about two times at a voltage of 1 V compared to the conventional electronic component.

(38) FIG. 4 is a cross-sectional view illustrating an electronic component **200** according to some example embodiments. Hereinafter, different content from the above-described example embodiments will be mainly described.

(39) Referring to FIG. 4, the electronic component **200** includes the first and second electrodes **111** and **112** apart from each other, a dielectric layer **220** provided between the first and second electrodes **111** and **112**, and a leakage current reducing layer **230** provided between the first electrode **111** and the dielectric layer **220**. Because the first and second electrodes **111** and **112** have

been described above, a description thereof will not be given herein.

(40) The dielectric layer **220** is similar to the first and second dielectric layers **121** and **122** in FIG. **1** described above. For example, the dielectric layer **220** may include a metal oxide represented by $M_{\text{sub}.x}O_{\text{sub}.y}$ (x and y are natural numbers). M may be one metal element selected from Ca, Sr, Ba, Sc, Y, La, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb and/or Lu. The dielectric layer **220** may have a single-layer structure including a single material layer or a multi-layer structure in which a plurality of material layers are stacked.

(41) The leakage current reducing layer **230** may be provided between the first electrode **111** and the dielectric layer **220**. The leakage current reducing layer **230** may include a metal oxide represented by $La_{\text{sub}.x}M'_{\text{sub}.y}O_{\text{sub}.z}$. M' may be one metal element selected from Ca, Sr, Ba, Sc, Y, La, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb and/or Lu. Such metal oxides may have paraelectric properties.

(42) The leakage current reducing layer **230** may be provided between the first electrode **111** and the dielectric layer **220**. The leakage current reducing layer **230** may include a metal oxide represented by $La_{\text{sub}.x}M'_{\text{sub}.y}O_{\text{sub}.z}$. M' may be one metal element selected from Ca, Sr, Ba, Sc, Y, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb and/or Lu. Such metal oxides may have paraelectric properties.

(43) FIG. **5** is a cross-sectional view illustrating an electronic component **300** according to some example embodiments.

(44) Referring to FIG. **5**, the electronic component **300** includes the first and second electrodes **112** and **112** apart from each other, a dielectric layer **320** provided between the first and second electrodes **111** and **112**, and a leakage current reducing layer **330** provided between the second electrode **112** and the dielectric layer **320**. Because the first and second electrodes **111** and **112**, the dielectric layer **320**, and the leakage current reducing layer **330** may be substantially similar to the first and second electrodes **111** and **112**, the dielectric layer **220**, the leakage current reducing layer **230**, as described above, a description thereof will not be given herein.

(45) According to another aspect, a semiconductor apparatus may be provided. The semiconductor apparatus may have a form in which a field effect transistor and a capacitor are electrically connected to each other, and the capacitor may be at least one of the aforementioned electronic component device **100**, **200**, or **300**. The semiconductor apparatus may have a memory characteristic, and may be, for example, DRAM. However, this is merely an example, the semiconductor apparatus is not limited thereto.

(46) FIG. **6** is a view of a semiconductor apparatus **D1** according to an example embodiment.

(47) Referring to FIG. **6**, the semiconductor apparatus **D1** may include a field effect transistor **10** and a capacitor **400** electrically connected to each other by the contact **20**. The field effect transistor **10** may include a substrate **11** including a channel **11c** and a gate electrode **12b** arranged to face the channel **11c**. A dielectric layer **12a** may be provided between the substrate **11** and the gate electrode **12b**.

(48) The substrate **11** may include a semiconductor material. The substrate **11** may include, for example, a semiconductor material such as silicon (Si), germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), and/or indium phosphide (InP), and may be used after being modified into various forms such as silicon on insulator (SOI).

(49) The substrate **11** may include a source **11a**, a drain **11b**, and a channel **11c** electrically connected to the source **11a** and the drain **11b**. The source **11a** may be electrically connected to or in contact with one side of the channel **11c**, and the drain **11b** may be electrically connected to or in contact with the other side of the channel **11c**. For example, the channel **11c** may be defined as a substrate area between the source **11a** and the drain **11b** in the substrate **11**.

(50) The source **11a**, the drain **11b**, and/or the channel **11c** may be each independently formed by implanting impurities into different areas of the substrate **11**, and in this case, the source **11a**, the

channel **11c**, and the drain **11b** may include a substrate material as a base material.

(51) The source **11a** and the drain **11b** may be formed of and/or include a conductive material. Each of the source **11a** and the drain **11b** may include, for example, a metal, a metal compound, and/or a conductive polymer. The conductive material may be, for example, an electrode and/or electrode contact.

(52) The channel **11c** may be implemented as a material layer (thin film) (not shown) separate from the substrate **11**. In this case, for example, the channel **11c** may include an oxide semiconductor, a nitride semiconductor, an oxynitride semiconductor, a two-dimensional (2D) material, a quantum dot, and/or an organic semiconductor, as well as a semiconductor material such as Si, Ge, SiGe, III-V, and/or the like. For example, the oxide semiconductor may include InGaZnO and/or the like, the 2D material may include transition metal dichalcogenide (TMD) and/or graphene, and the quantum dot may include a colloidal quantum dot (QD), a nanocrystal structure, and/or the like.

(53) The gate electrode **12b** may be arranged on the substrate **11** to face the channel **11c** while being apart from the substrate **11**. The gate electrode **12b** may have, for example, conductivity of 1 Mohm/square or less. The gate electrode **12b** may include a conductive material such as a metal, metal nitride, metal carbide, and/or polysilicon. For example, the metal may include aluminum (Al), tungsten (W), molybdenum (Mo), titanium (Ti), and/or tantalum (Ta), and the metal nitride film may include a titanium nitride film (TiN film), and/or a tantalum nitride film (Ta_N film). The metal carbide may be a metal carbide doped with (or containing) aluminum and/or silicon, and may include TiAlC, TaAlC, TiSiC, and/or TaSiC as some specific examples.

(54) The gate electrode **12b** may have a structure in which a plurality of materials are stacked. For example, the gate electrode **12b** may be and/or include a stacked structure of a metal nitride layer/metal layer such as TiN/Al, and/or a stacked structure of a metal nitride layer/metal carbide layer/metal layer such as TiN/TiAlC/W. The gate electrode **12b** may include a TiN and/or Mo, and the above example may be used in variously modified forms.

(55) A gate insulating layer **12a** may be further provided between the substrate **11** and the gate electrode **12b**. The gate insulating layer **12a** may include a paraelectric material or a high-k dielectric material. The gate insulating layer **12a** may include a material having a dielectric constant of about 20 to 70. For example, the gate insulating layer **12a** may include silicon oxide, silicon nitride, aluminum oxide, hafnium oxide, zirconium oxide, and/or the like, and/or a 2D insulator such as hexagonal boron nitride (h-BN).

(56) For example, the gate insulating layer **12a** may include at least one of silicon oxide (SiO₂), silicon nitride (Si₃N₄), and/or the like, and/or may include hafnium oxide (HfO₂), hafnium silicon oxide (HfSiO₄), lanthanum oxide (La₂O₃), lanthanum aluminum oxide (LaAlO₃), zirconium oxide (ZrO₂), hafnium zirconium oxide (HfZrO₂), zirconium silicon oxide (ZrSiO₄), tantalum oxide (Ta₂O₅), titanium oxide (TiO₂), strontium titanium oxide (SrTiO₃), yttrium oxide (Y₂O₃), aluminum oxide (Al₂O₃), red scandium tantalum oxide (PbSc_{0.5}Ta_{0.5}O₃), red zinc niobate (PbZnNbO₃), and/or the like. In addition, the gate insulating layer **12a** may include metal nitride oxides such as aluminum oxynitride (AlON), zirconium oxynitride (ZrON), hafnium oxynitride (HfON), lanthanum oxynitride (LaON), yttrium oxynitride (YON), and/or the like, silicate such as ZrSiON, HfSiON, YSiON, LaSiON, and/or the like, and/or aluminate such as ZrAlON, HfAlON, and/or the like. However, these are merely examples, and the gate insulating layer **12a** may be not limited thereto. The gate insulating layer **12a** may constitute a gate stack together with the gate electrode **12b**.

(57) The capacitor **400** may be at least one of the electronic component **100**, **200**, and/or **300** according to the above-described embodiments. FIG. 6 shows an example case in which the capacitor **400** has the structure of the semiconductor device **100** shown in FIG. 1. In this case, first and second dielectric layers **421** and **422** are provided between first and second electrodes **411** and **412**, and a leakage current reducing layer **430** including a metal oxide represented by

La.sub.x'M'.sub.yO.sup.z' is provided between the first and second dielectric layers **421** and **422**.

However, this is only an example, and the capacitor **400** may have, as described above, the structure of the semiconductor devices **200** and/or **300** shown in FIG. 4 or FIG. 5. Because the capacitor **400** has been described above, a detailed description thereof will not be given herein.

(58) The field effect transistor **10** and the capacitor **400** may be electrically connected to each other by the contact **20**. For example, one of the first and second electrodes **411** and **412** of the capacitor **400** and one of the source and drain **11a** and **11b** of the field effect transistor **10** may be electrically connected to each other by the contact **20**. The contact **20** may include a conductive material, for example, tungsten, copper, aluminum, polysilicon, and/or the like. The arrangement of the capacitor **400** and the field effect transistor **10** may vary. For example, the capacitor **400** may be arranged on the substrate **11** or may be embedded in the substrate **11**.

(59) FIG. 7 is a view of a semiconductor apparatus **D10** according to some example embodiments. The semiconductor apparatus **D10** shown in FIG. 7 has a structure in which a plurality of capacitors **500** and a plurality of field effect transistors are repeatedly arranged.

(60) Referring to FIG. 7, the semiconductor apparatus **D10** may include a field effect transistor having the substrate **11'** including a source **11a**, a drain **11b**, a channel **11c**, and the gate stack **12**, the contact structure **20'** arranged on the substrate **11'** so as not to overlap the gate stack **12**, and a capacitor **500** arranged on the contact structure **20'**, and may further include the bit line structure **13** electrically connecting the plurality of field effect transistors.

(61) FIG. 7 illustrates an example of the semiconductor apparatus **D10** in which both contact structures **20'** and the capacitors **500** are repeatedly arranged in X and Y directions, but the semiconductor apparatus **D10** may be not limited thereto. For example, the contact structures **20'** may be arranged in the X direction and the Y direction, and the capacitors **500** may be arranged in a hexagonal shape such as a honeycomb structure.

(62) FIG. 8 is a cross-sectional view of the semiconductor apparatus **D10** taken along line A-A' of FIG. 7.

(63) Referring to FIG. 8, the substrate **11'** may have a shallow trench isolation (STI) structure including a device isolation layer **14**. The substrate **11'** may be the same and/or substantially similar to the above-described substrate **11**. The device isolation layer **14** may be a single layer formed of one type of insulating layer, or a multilayer formed of a combination of two or more types of insulating layers. The device isolation layer **14** may include a device isolation trench **14T** in the substrate **11'**, and the device isolation trench **14T** may be filled with an insulating material. The insulating material may include, but is not limited to, fluoride silicate glass (FSG), undoped silicate glass (USG), boro-phospho-silicate glass (BPSG), phospho-silicate glass (PSG), flowable oxide (FOX), plasma enhanced tetra-ethyl-ortho-silicate (PE-TEOS), tonen silazene (TOSZ), and/or the like.

(64) The substrate **11'** may further include an active area **AC** defined by the device isolation layer **14** and a gate line trench **12T** that is parallel to an upper surface of the substrate **11'** and is arranged to extend in the X direction. The active area **AC** may have a relatively long island shape having a short axis and a long axis. The long axis of the active area **AC** may be arranged in a K direction parallel to the upper surface of the substrate **11'** as exemplarily illustrated in FIG. 7. The gate line trench **12T** may be arranged to cross the active area **AC** at a certain depth from the upper surface of the substrate **11'** or may be arranged in the active area **AC**. The gate line trench **12T** may also be arranged inside the device isolation trench **14T**, and the gate line trench **12T** inside the device isolation trench **14T** may have a lower bottom surface than the gate line trench **12T** of the active area **AC**.

(65) A first source/drain **11'ab** and a second source/drain **11''ab** may be arranged in an upper portion of the active area **AC** located at both sides of the gate line trench **12T**. The first source/drain **11'ab** and/or the second source/drain **11''ab** may be the same and/or substantially similar to the above described source **11a** and/or drain **11b**.

(66) The gate stack **12** may be arranged inside the gate line trench **12T**. For example, the gate stack **12** may include the gate insulating layer **12a**, the gate electrode **12b**, and/or a gate capping layer **12c** sequentially arranged in the gate line trench **12T**. The gate insulating layer **12a** and the gate electrode **12b** may refer to the above description, and the gate capping layer **12c** may include silicon oxide, silicon oxynitride, and/or silicon nitride. The gate capping layer **12c** may be arranged on the gate electrode **12b** to fill the remaining portion of the gate line trench **12T**.

(67) The bit line structure **13** may be arranged on the first source/drain **11'ab**. The bit line structure **13** may be arranged to be parallel to the upper surface of the substrate **11'** and extend in the Y direction. The bit line structure **13** is electrically connected to the first source/drain **11'ab**, and may include a bit line contact **13a**, a bit line **13b**, and a bit line capping layer **13c** sequentially stacked on the substrate **11'**. For example, the bit line contact **13a** may include polysilicon, the bit line **13b** may include a metal material, and the bit line capping layer **13c** may include an insulating material such as silicon nitride or silicon oxynitride. In FIG. 8, a case in which the bit line contact **13a** has a bottom surface at the same level as the upper surface of the substrate **11'** is exemplarily shown. However, the bit line contact **13a** may extend from the upper surface of the substrate **11'** to the inside of a recess (not shown) formed to a certain depth, so that the bottom surface of the bit line contact **13a** may be lower than the upper surface of the substrate **11'**.

(68) The bit line structure **13** may further include a bit line intermediate layer (not shown) between the bit line contact **13a** and the bit line **13b**. The bit line intermediate layer may include a metal silicide such as tungsten silicide, and/or a metal nitride such as tungsten nitride. In addition, a bit line spacer (not shown) may be further formed on a sidewall of the bit line structure **13**. The bit line spacer may have a single-layer structure or a multi-layer structure, and may include an insulating material such as silicon oxide, silicon oxynitride, or silicon nitride. In addition, the bit line spacer may further include an air space (not shown).

(69) The contact structure **20'** may be arranged on the second source/drain **11''ab**. The contact structure **20'** and the bit line structure **13** may be arranged on different sources/drains on the substrate **11'**. The contact structure **20'** may have a structure in which a lower contact pattern (not shown), a metal silicide layer (not shown), and an upper contact pattern (not shown) are sequentially stacked on the second source/drain **11''ab**. In addition, the contact structure **20'** may further include a barrier layer (not shown) surrounding a side surface and a bottom surface of the upper contact pattern. For example, the lower contact pattern may include polysilicon, the upper contact pattern may include a metal material, and the barrier layer may include a conductive metal nitride.

(70) The capacitor **500** may be electrically connected to the contact structure **20'** and arranged on the substrate **11'**. The capacitor **500** may be one of the electronic components **100**, **200**, and/or **300**, and/or the capacitor **400**, according to the above-described embodiments. FIG. 8 shows an example case in which the capacitor **500** has the structure of the semiconductor device **100** shown in FIG. 1.

(71) For example, the capacitor **500** may include a first electrode **511** electrically connected to the contact structure **20'**, a second electrode **512** provided to be apart from the first electrode **511**, first and second dielectric layers **521** and **522** provided between the first and second electrodes **511** and **512**, a leakage current reducing layer **530** including a metal oxide represented by $\text{La.sub.xM'.sub.yO.sub.z'}$ provided between the first and second dielectric layers **521** and **522**. However, this is only an example, and the capacitor may have the structure of the semiconductor devices **200** and **300** shown in FIG. 4 or FIG. 5.

(72) The interlayer insulating layer **15** may be further arranged between the capacitor **500** and the substrate **11'**. The interlayer insulating layer **15** may be arranged in a space between the capacitor **500** and the substrate **11'** in which other structures are not arranged. For example, the interlayer insulating layer **15** may be arranged to cover wiring and/or electrode structures such as the bit line structure **13** on the substrate **11'**, the contact structure **20'**, and the gate stack **12**. For example, the interlayer insulating layer **15** may surround a wall of the contact structure **20'**. The interlayer

insulating layer **15** may include a first interlayer insulating layer **15a** surrounding the bit line contact **13a** and a second interlayer insulating layer **15b** covering side surfaces and/or upper surface of the bit line **13b** and the bit line capping layer **13c**.

(73) The first electrode **511** of the capacitor **500** may be arranged on the interlayer insulating layer **15**, specifically, on the second interlayer insulating layer **15b**. In addition, when a plurality of capacitors **500** are arranged, bottom surfaces of a plurality of first electrodes **511** may be separated by an etch stop layer **16**. For example, the etch stop layer **16** may include an opening **16T**, and a bottom surface of a first electrode **100** of the capacitor **500** may be arranged in the opening **16T**.

(74) The first electrode **511** may have a cylinder shape with a closed bottom and/or a cup shape, as shown in FIG. **8**, and/or, in another example, as in a capacitor **500'** illustrated in FIG. **9**, the first electrode **511** may have a pillar shape such as a cylinder, a square pillar, and/or a polygonal pillar extending in a vertical direction (Z direction). The capacitor **500** may further include a support portion (not shown) for preventing (e.g., mitigating the potential for) the first electrode **511** from being tilted and/or collapsed, and the support portion may be arranged on a sidewall of the first electrode **511**.

(75) The above-described semiconductor apparatus **D10** may be manufactured with reference to a conventional method known in the art. For example, the semiconductor apparatus **D10** may be manufactured by including operations of i) to xvi) below.

(76) i) Forming the device isolation trench **14T** in the substrate **11'**, and forming the device isolation layer **14** in the device isolation trench **14T** (Defining the active area AC of a substrate **102** by the device isolation layer **14** and/or the device isolation trench **14T**),

(77) ii) Filling the inside of the isolation trench **14T** with an insulating material,

(78) iii) Implanting impurity ions into the substrate **11'** to form the first source/drain **11'ab** and the second source/drain **11''ab** in an upper area of the active area AC,

(79) iv) Forming the gate line trench **12T** in the substrate **11'**,

(80) v) Forming the gate insulating layer **12a**, the gate electrode **12b**, and the gate capping layer **12c** inside the gate line trench **12T**,

(81) vi) Forming the first interlayer insulating layer **15a** on the substrate **11'**, and forming an opening (not shown) exposing an upper surface of the first source/drain **11'ab**,

(82) vii) Forming the bit line structure **13** electrically connected to the first source/drain **11'ab** on the opening of vi),

(83) viii) Forming the second interlayer insulating layer **15b** covering an upper surface and a side surface of the bit line structure **13**,

(84) ix) Forming an opening (not shown) in the first and second interlayer insulating layers **15a** and **15b** to expose an upper surface of the second source/drain **11''ab**,

(85) x) Forming the contact structure **20'** electrically connected to the second source/drain **11''ab** on the opening of ix),

(86) x i) Forming the etch stop layer **16** and a mold layer (not shown) on the second interlayer insulating layer **15b** and the contact structure **20'**,

(87) x ii) Forming an opening (not shown) in the etch stop layer **16** and the mold layer (not shown) to expose an upper surface of the contact structure **20'**,

(88) x iii) Forming the first electrode **511** to cover an inner wall (to cover bottom and side surfaces) of the opening of x ii),

(89) x iv) Removing the mold layer (not shown),

(90) x v) Forming the first dielectric layer **521**, a leakage current reducing layer **530**, and the second dielectric layer **522** on the first electrode **511**, and

(91) x vi) Forming the second electrode **512** on the second dielectric layer **522**.

(92) The type and/or order of each operation described above is not limited, and may be appropriately adjusted, and some may be omitted and/or added. In addition, a deposition process, a patterning process, an etching process, etc. known in the art may be used to form the components in

each operation. For example, an etch-back process may be applied when forming an electrode. In operation v), the gate electrode **12b** may be formed by forming a conductive layer on the gate insulating layer **12a** and then removing an upper portion of the conductive layer by a certain height through an etch-back process. In addition, in operation x iii), the first electrode **511** may also be manufactured in a structure including a plurality of first electrodes **511** by forming an electrode to cover all of an upper surface of the mold layer and the bottom and side surfaces of the opening and then removing a portion of the electrode on the upper surface of the mold layer by an etch-back process. As another example, a planarization process may be applied. For example, in operation v), the gate capping layer **12c** may be formed by filling the remaining portion of the gate line trench **12T** with an insulating material and then planarizing the insulating material until the upper surface of the substrate **11'** is exposed.

(93) According to another aspect, the above-described electronic component **100**, **200**, and/or **300** and/or the semiconductor apparatuses **D1** and/or **D10** may be applied to various electronic devices. For example, the above-described electronic component **100**, **200**, and/or **300** and/or the semiconductor apparatuses **D1** and/or **D10** may be applied as logic devices or memory devices in various electronic devices. For example, the electronic component **100**, **200**, and/or **300** and the semiconductor apparatuses **D1** and **D10** may be used for arithmetic operations, program execution, and temporary data retention in electronic devices such as mobile devices, computers, laptops, sensors, network devices, or neuromorphic devices. A semiconductor device and a semiconductor apparatus according to embodiments may be useful for electronic devices in which the amount of data transmission is large and data transmission is continuously performed.

(94) FIGS. **10** and **11** are conceptual diagrams schematically illustrating a device architecture applicable to an electronic device according to some example embodiments.

(95) Referring to FIG. **10**, a device architecture **1000** may include a memory unit **1010**, an arithmetic logic unit (ALU) **1020**, and a control unit **1030**. The memory unit **1010**, the ALU **1020**, and the control unit **1030** may be electrically connected to each other. For example, the device architecture **1000** may be implemented as a single chip including the memory unit **1010**, the ALU **1020**, and the control unit **1030**. In more detail, the memory unit **1010**, the ALU **1020**, and the control unit **1030** may be interconnected by a metal line in an on-chip to directly communicate with each other. The memory unit **1010**, the ALU **1020**, and the control unit **1030** may be integrated monolithically on one substrate to configure one chip. An input/output device **2000** may be connected to the device architecture **1000**. In addition, the memory unit **1010** may include both a main memory and a cache memory. This device architecture **1000** may be an on-chip memory processing unit. The memory unit **1010**, the ALU **1020**, and/or the control unit **1030** may each independently include the aforementioned semiconductor device.

(96) Referring to FIG. **11**, a cache memory **1510**, an ALU **1520**, and a control unit **1530** may constitute a central processing unit (CPU) **1500**, and the cache memory **1510** may be formed of static random access memory (SRAM). Separately from the CPU **1500**, a main memory **1600** and an auxiliary storage **1700** may be provided. The main memory **1600** may be dynamic random access memory (DRAM) and may include the aforementioned semiconductor device. In some cases, the device architecture may be implemented in a form in which computing unit devices and memory unit devices are adjacent to each other in a single chip without distinction of sub-units.

(97) According to the above example embodiments, by providing a leakage current reducing layer including $\text{La.sub.x'M'.sub.yO.sub.z'}$ metal oxide in a dielectric layer, electronic component capable of increasing capacitance while reducing a leakage current may be realized. Such an electronic component may be applied to a semiconductor apparatus such as DRAM, and an electronic device such as a mobile device, a computer, a laptop, a sensor, a network device, and a neuromorphic device. Although some example embodiments have been described above, this is merely examples, and various modifications are possible therefrom by one of ordinary skill in the art.

(98) It should be understood that the example embodiments described herein should be considered

in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

Claims

1. An electronic component comprising: a first electrode; a second electrode apart from the first electrode; and a dielectric structure between the first electrode and the second electrode, the dielectric structure including a dielectric layer including a metal oxide represented by $M_{x'}O_{y'}$, where M is at least one of Ca, Sr, Ba, Sc, Y, La, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, or Lu, and a leakage current reducing layer including a metal oxide represented by $La_{x''}M'_{y''}O_{z''}$, where M' is at least one of Ca, Sr, Ba, Sc, Y, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, or Lu, wherein a thickness of the leakage current reducing layer is within a range of 0.1 Å to 4.5 Å.
2. The electronic component of claim 1, wherein the dielectric layer comprises first and second dielectric layers, and the leakage current reducing layer is between the first dielectric layer and the second dielectric layer.
3. The electronic component of claim 1, wherein the leakage current reducing layer is between the first electrode and the dielectric layer.
4. The electronic component of claim 1, wherein the leakage current reducing layer is between the second electrode and the dielectric layer.
5. The electronic component of claim 1, wherein a total thickness of the dielectric structure is about 50 Å or less.
6. The electronic component of claim 1, wherein the dielectric layer has at least one of a single-layer structure.
7. The electronic component of claim 1, wherein the dielectric layer has a multi-layer structure in which different materials are stacked.
8. The electronic component of claim 1, wherein each of the first and second electrodes comprises at least one of W, TaN, TiN, RuO_{1-3} , NbN, Sc, Al, Mo, MON, Pd, Pt, Sn, La, or Ru.
9. The electronic component of claim 1, wherein one of the first and second electrodes comprises a semiconductor material.
10. A semiconductor apparatus comprising: a field effect transistor; and a capacitor electrically connected to the field effect transistor, the capacitor comprising a first electrode, a second electrode apart from the first electrode, and a dielectric structure between the first electrode and the second electrode, the dielectric structure including a dielectric layer including a metal oxide represented by $M_{x'}O_{y'}$, where M is at least one of Ca, Sr, Ba, Sc, Y, La, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, and Lu, and a leakage current reducing layer including a metal oxide represented by $La_{x''}M'_{y''}O_{z''}$, where M' is at least one of Ca, Sr, Ba, Sc, Y, Ti, Zr, Hf, Nb, Ta, Ce, Pr, Nd, Gd, Dy, Yb, and Lu, wherein a thickness of the leakage current reducing layer is within a range of 0.1 Å to 4.5 Å.
11. The semiconductor apparatus of claim 10, wherein the field effect transistor comprises: a channel layer between a source and a drain; a gate layer on the channel layer; and a gate electrode on the gate layer.
12. The semiconductor apparatus of claim 10, wherein the dielectric layer comprises first and second dielectric layers, and the leakage current reducing layer is between the first dielectric layer and the second dielectric layer.
13. The semiconductor apparatus of claim 10, wherein the leakage current reducing layer is

between the dielectric layer and at least one of the first electrode or the second electrode.

14. The semiconductor apparatus of claim 10, wherein a total thickness of the dielectric structure is about 50 Å or less.

15. The semiconductor apparatus of claim 10, wherein each of the first and second electrodes comprises at least one of W, TaN, TiN, RuO.sub.1-3, NbN, Sc, Al, Mo, MON, Pd, Pt, Sn, La, or Ru.

16. An electronic device comprising a plurality of the semiconductor apparatus of claim 10.
